

Sebacic acid ($\text{HOOC}-(\text{CH}_2)_8-\text{COOH}$) is a naturally occurring dicarboxylic acid found in urine. A 200 mg sample of sebacic acid crystals would have the highest solubility in a dilute aqueous solution of:

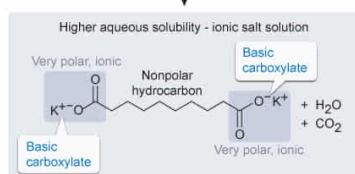
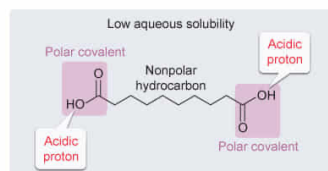
- ☐ A. a weak acid. [20%]
☒ B. a weak base. [58%]
☐ C. a neutral ionic salt. [8%]
☐ D. distilled water. [13%]

Correct

59% answered correctly

Explanation:

Effect of pH on the solubility of an acidic compound



The aqueous solubility of an acidic compound can often be increased by raising the pH of the solution (adding a base).

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Solubility refers to the maximum amount of a **solute** (ie, a dissolved compound) that can dissolve per unit volume of a **solvent** (ie, the solution medium) at a given temperature. Solubility depends on the ability of the solute and solvent molecules to interact through complementary, noncovalent **intermolecular forces** that form attractions between the molecules. Solute molecules that can associate with solvent molecules through **similar types of intermolecular forces** tend to be **attracted to the solvent**, and thus are **more soluble** in it.

The structure of sebacic acid has polar covalent C=O, C—O, and O—H bonds that readily interact with water via **dipole-dipole** and **hydrogen bonding** interactions. Nevertheless, the solubility of sebacic acid in water is still very limited due to the nonpolar C—C and C—H bonds of the central hydrocarbon chain, which is **hydrophobic** and repels water molecules.

Because **bases neutralize acids**, the solubility of sebacic acid can be increased by adding a base to **deprotonate** the **carboxylic acid** groups. The resulting **acid-base neutralization reaction** converts the carboxylic acid groups into ionic carboxylate salts, which have much higher aqueous solubility because they are charged and highly polar. The **added charge** increases the **attraction to water** and **overcomes the repulsion** from the hydrocarbon chain.

Therefore, a sample of sebacic **acid crystals** would have the **highest solubility** in a dilute **basic solution**.

(Choice A) Sebacic acid is a **weak acid**, and adding another weak acid will not increase solubility because two weak acids will not neutralize or ionize each other.

(Choice C) A neutral ionic salt is neither basic nor acidic (ie, does not affect the pH of the solution) and will not affect the solubility of sebacic acid.

(Choice D) Distilled water has a **pH near 7** and will not affect the ionization of sebacic acid.

Educational objective:

The solubility of a compound being dissolved depends on its ability to form complementary intermolecular interactions with solvent molecules. Acid-base neutralization reactions can be used to increase the amount of an acidic (or basic) solute that will dissolve in a solvent by forming more soluble ionic salts.